THE LOCUS WHERE A BASE CHANGE IS SURJECTIVE AND FLAT IS OPEN

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 11750-11789, immediately following the statement at 11728-11748.

Editorial note. Five cross-chapter pointers (Morphisms of Spaces, Properties of Spaces) plus one same-chapter lemma are recorded in assembly.external_references.

Editorial note. The comparison square of point sets is displayed verbatim inside the span.

Allowed standard prerequisites (editorial).

- The flatness locus for the same diagram. chapter lemma lemma-where-flat, cited in the proof. Supplies the open subspace of Z over which the base change of f is flat, which is half of the conclusion.
- Open subspaces and factorisation through them. Properties of Spaces, lemmas spaces-properties-lemma-open-subspaces and spaces-properties-lemma-factor-through-open-subspace, cited in the proof. Realise the constructed point sets as open subspaces and turn pointwise descriptions into universal properties.
- Points of a base change square. Properties of Spaces, lemma spaces-properties-lemma-points-cartesian, cited at proof lines 11777-11778. Identifies the fibres in the comparison square of point sets used for surjectivity.
- Openness of fppf morphisms and permanence of finite presentation. Morphisms of Spaces, lemmas spaces-morphisms-lemma-fppf-open and spaces-morphisms-lemma-finite-presentation-permanence, cited in the proof. Control the images of the relevant subsets under the morphisms of the diagram.

Source reference. Original lemma 05XB: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11687–11687. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05WT: focused-seeds/geometry/S9-spaces-where-surjective-flat-spaces/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 042S: focused-seeds/geometry/S9-spaces-where-surjective-flat-spaces/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 03IE: focused-seeds/geometry/S9-spaces-where-surjective-flat-spaces/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03BZ: focused-seeds/geometry/S9-spaces-where-surjective-flat-spaces/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03H4: focused-seeds/geometry/S9-spaces-where-surjective-flat-spaces/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Consider a commutative diagram*

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & {\scriptstyle p}\searrow \quad \swarrow{\scriptstyle q} & \\ & Z & \end{array}$$

of algebraic spaces. Assume that

(1) *p is locally of finite presentation,*
(2) *p is flat,*
(3) *p is closed,*
(4) *q is locally of finite type, and*
(5) *q is closed.*

Then there exists an open subspace $W \subset Z$ such that a morphism $Z' \to Z$ factors through W if and only if the base change $f_{Z'} : X_{Z'} \to Y_{Z'}$ is surjective and flat.

Proof. By Lemma 05XB we may assume that f is flat. Note that f is locally of finite presentation by Morphisms of Spaces, Lemma 05WT. Hence f is open, see Morphisms of Spaces, Lemma 042S. Let $W \subset Z$ be the open subspace (see Properties of Spaces, Lemma 03BZ) with underlying set of points

$$|W| = |Z| \setminus |q| \left(|Y| \setminus |f|(|X|) \right).$$

in other words for $z \in |Z|$ we have $z \in |W|$ if and only if the whole fibre of $|Y| \to |Z|$ over z is in the image of $|X| \to |Y|$. Since q is closed this set is open in $|Z|$. The morphism $X_W \to Y_W$ is surjective by construction. Finally, suppose that $X_{Z'} \to Y_{Z'}$ is surjective. In order to show that $Z' \to Z$ factors through W it suffices to show that $|Z'| \to |Z|$ has image contained in $|W|$, see Properties of Spaces, Lemma 03IE. Hence it suffices to prove the result when Z' is the spectrum of a field. Denote $z \in |Z|$ the image of $|Z'| \to |Z|$. By Properties of Spaces, Lemma 03H4 in the commutative diagram

$$\begin{array}{ccc} |X_{Z'}| & \longrightarrow & |Y_{Z'}| \\ \downarrow & & \downarrow \\ |p|^{-1}(\{z\}) & \longrightarrow & |q|^{-1}(\{z\}) \end{array}$$

the vertical arrows are surjective. It follows that $z \in |W|$ as desired. □